

# Tanay Das

AMD-Xilinx (2025-Present)  
MTECH – IIT Gandhinagar, Gujarat

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## EDUCATION

**Indian Institute of Technology, Gandhinagar**  
M.Tech, Microelectronics VLSI, Electrical Engineering  
*Advisor: Prof. Sandip Lashkare*  
CPI: 9.06

**Gujrat, India**  
July 2023 - June 2025

**Sikkim Manipal Institute of Technology**  
B.Tech, Electronics and Communication Engineering  
CPI: 9.7

**Sikkim, India**  
July 2019 - June 2023

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## PUBLICATIONS

### JOURNAL PUBLICATIONS

1. **ESD Protection with Sub-3  $\Omega$  Dynamic Resistance for Neuromodulation System-on-Chip** - Tanay Das, Naef Ahmad, Laxmeesha Somappa, Sandip Lashkare. – **JEDS** [Link](#) 2025

### CONFERENCE PUBLICATIONS

1. **Design of ESD Protection with on-chip diodes for 65nm CMOS High Voltage Neurostimulators** - Tanay Das, Laxmeesha Somappa, Sandip Lashkare. – **ICEE** [Link](#) 2025
2. **Hybrid Software-FPGA RRAM-crossbar Array Platform For In-Memory Computing With Bit-Slicing Architecture (Best Paper Award)** - Udisha Singh, Tanay Das, et. al – **ICEE** [Link](#) 2025
3. **TCAD-Guided Design of a Compact ESD Protection Circuit for Neurostimulators in CMOS 65nm Technology** - Navin Maheshwari, Tanay Das, et. al. – **ICEE** [Link](#) 2025
4. **Enhanced ESD Protection Techniques for 10V Neurostimulator Circuits in 65nm CMOS Technology** - Tanay Das, Naef Ahmad, Laxmeesha Somappa, Sandip Lashkare. – **EDTM** [Link](#) 2025
5. **Dynamic Resistance Reduction Methods for Voltage Clamp Lowering to Enhance GGNMOS ESD Protection** - Tanay Das, Madhav Pathak, Sandip Lashkare.– **VDAT** [Link](#) 2023
6. **Synthesis and Characterization of MXene from MAX phase** - Arijit Ghosh, Himangshu Pal, Tanay Das, Somenath Chatterjee, Amalendu Das. **Materialstoday: Proceedings** [Link](#) 2022

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## RESEARCH INTERESTS

Emerging Non-volatile Memories: RRAM, Neuromorphic Computing, Analog and Mixed Signal IC Design

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## INDUSTRY EXPERIENCE

- **AMD, Hyderabad** Jul, 2025 - Present  
*Silicon Design Engineer*
  - o Performed regression analysis and debugging for RTL verification flows.
  - o Executed and analysed VHDL/Verilog testcases using XSIM and industry simulators Xcelium, Questa, Riviera.
  - o Developed and tested SystemVerilog **\$deposit**-based implementations for signal forcing and verification use-cases.

- **StarIC, Canada** May, 2024 - Jul, 2024  
*Photonics IC Design Intern*  
 o Designed and simulated WDM EOM and DCF models for a 4-bit serial DAC using gdsfactory and python.
- **Space Application Centre (SAC), ISRO, Ahmedabad** Jan, 2023 - May, 2023  
*Research Intern*  
 o Design layout and analysis of Radiation Hardened digital library in 65 nm technology Node
  - Researched and implemented radiation-hardening techniques for digital circuits using 65 nm node.
  - Developed and implemented EDAC logic, reducing error rates by 90 and improving system resilience.
  - Utilized Cadence Virtuoso for circuit analysis and design.

## MTECH THESIS

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- **ESD protection/clamping circuits for brain-machine interface (BMI), IIT Gandhinagar** Jan, 2024 - Jun 2025  
*Advisor: Prof. Sandip Lashkare*
    - Designed and analyzed ESD protection circuits for 180 nm technology, optimizing layouts for reliability.
    - Enhanced ESD Protection Techniques for 10V Neurostimulator Circuits in 65nm CMOS Technology

## COURSE PROJECTS

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- **Design a 2-stage OPAMP with Miller Compensation, IIT Gandhinagar** Jan, 2024 - Apr, 2024  
*Instructor: Prof. Nihar Ranjan Mahapatra*
    - Designed a 2-stage OPAMP with Miller Compensation, Differential Gain >80 dB, CMRR >100dB, ICMR: 0.8 – 1.8V, Max. Output Swing: 0.8V, GBW >10MHz, PM >65deg, and Power dissipation <0.5mW with a load of 1 pF.
  - **Design a Folded Cascode Operational Amplifier with single-ended output, IIT Gandhinagar** Jan, 2024 - Apr, 2024  
*Instructor: Prof. Nihar Ranjan Mahapatra*
    - Specifications: Differential Gain >90 dB, CMRR >100dB, ICMR: 0.8 – 1.8V, Max. Output Swing: 800mV, Power <20uW, GBW >10MHz
  - **Design a band-gap reference circuit, IIT Gandhinagar** Jan, 2024 - Apr, 2024  
*Instructor: Prof. Nihar Ranjan Mahapatra*
    - Specifications: Temperature regulation: 10mV, Supply voltage regulation: 10 mV, Power budget: 50uW

## TEACHING EXPERIENCE

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- 2024-25 Semester 1 EE322 Analog and Mixed Signal Circuits (Tutor)
  - 2023-24 Semester 2 ES204 Digital System (Tutor)

## TECHNICAL SKILLS

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- **Design Tools:** XILINX VIVADO, Cadence Virtuoso, NGSpice, Magic, L<sup>A</sup>T<sub>E</sub>X.
  - **Programming/Scripting:** Verilog, PYTHON, MATLAB.

## SELECTED AWARDS AND RECOGNITIONS

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- Recipient of URAM 2.0 Scholarship, Micron [2024]
  - Winner of the "Sky 130 mixed-signal design Marathon", organized by FOSSEE IIT Bombay [2022]
  - Winner of the "Capture The Bug-Design Verification Hackathon," organized by the NIELIT [2022]
  - Certificate of Academic Excellence (Gold Medalist), Sikkim Manipal Institute of Technology. [2021, 2022]